



Mentor **Link**

Hack it Project

Team Name: **Next Step**

Leader: Haneen Mohamed Elmetwally

Track: UI / UX Design

Team Members:

- Dina Mohamed Hamza - UI / UX Design
- Abdullah Sabri Fayed - AI & Machine Learning
- Hager Elmalah - Data Analysis
- Gamal Azzam - Flutter (Mobile App Development)
- Mohamed Elsayed - Flutter (Mobile App Development)
- Ahmed Qassem - Backend Development
- Bassim Ehab Mohamed - Backend Development

SDGs Fulfilled

Our platform directly contributes to several UN Sustainable Development Goals (SDGs). By offering students and fresh graduates access to mentors, AI-tailored learning roadmaps, and an interactive chatbot for technical guidance, we support:

- SDG 4 (Quality Education) through personalized, accessible skill development.
- We also advance SDG 8 (Decent Work and Economic Growth) by bridging the gap between academia and industry, enhancing employability and fostering innovation.
- Leveraging AI as a scalable solution aligns with SDG 9 (Industry, Innovation, and Infrastructure).
- Providing equal access to mentorship regardless of background fulfills SDG 10 (Reduced Inequalities).
- Finally, by connecting students, professionals, and institutions, the platform strengthens collaboration in line with SDG 17 (Partnerships for the Goals).

Problem Statement

Students and fresh graduates often struggle to transition from academic knowledge to practical, industry-relevant skills. Many lack access to experienced mentors who can provide guidance, career direction, and networking opportunities. Existing learning platforms offer content but fail to provide personalized, structured roadmaps tailored to individual goals. This leads to uncertainty, skill mismatches, and limited job opportunities. Additionally, students frequently face technical challenges without reliable, accessible support, which slows down their learning journey. Without a solution that combines mentorship, AI-driven learning paths, and instant technical assistance, a significant number of young talents risk being underprepared for the demands of the job market.

Proposed Solution

Students and fresh graduates often face challenges bridging the gap between academic knowledge and industry demands due to limited mentorship, unclear learning paths, and lack of technical support.

Our platform solves this by offering AI-driven roadmaps tailored to career goals, connecting users with experienced mentors for guidance, and providing an AI chatbot for instant technical answers. This all-in-one ecosystem equips young talents with the right skills, confidence, and support to become job-ready, fostering innovation and inclusive economic growth.

Value

- A unified platform combining human mentorship and AI technology to deliver practical, guided learning experiences.
- Personalized learning roadmaps tailored to each student's or graduate's career goals.
- Equal access to mentorship and career support, regardless of background or resources.

Benefits

- Smooth transition from education to the job market with confidence.
- Improved employability and entrepreneurship opportunities through relevant, in-demand skills.
- Instant technical support via the AI chatbot, saving time and boosting learning efficiency.
- Professional networking by connecting with mentors and industry experts.

Technologies Used

UI / UX Design

- Using Figma Software
 - The primary color, **#A9E139**, the greenish tone symbolizes personal and professional growth, aligning with the app's mission to guide students and graduates.
 - The secondary color, **#38ACDD**, blue is universally linked to trust, making users feel secure in the guidance and AI support they receive.
 - The Logo symbolizes connection & Mentorship, the two curved shapes look like links or paths coming together, symbolizing mentors and mentees.

Technologies Used

Mobile App Development (Flutter)

- We built the app with Flutter (Dart) using Cubit/BLoC for state management and Clean Architecture for scalability. Firebase powers cloud services, while Hive handles local storage. For backend integration, we use dio, and get_it manages dependency injection. go_router supports navigation, with flutter_screenutil ensuring responsiveness across devices. UI testing and previews are streamlined with device_preview.

Backend Development

- Backend built the platform using ASP.NET Core (.NET 9) with Entity Framework Core as the ORM and SQL Server (MSSQL) for the database. For authentication, implemented JWT tokens to ensure secure access. Postman was used for testing and validating the APIs.

Technologies Used

AI & Machine Learning

OpenAI API client

DeepSeek API – API for DeepSeek large language models

Sentence-transformers – Text, sentence embeddings

Cosine Similarity – Vector similarity metric (for semantic search & embeddings)

RAG & Vector Database :

LlamaIndex – Framework for building LLM-powered apps

(LlamaIndex-llms-openai) – OpenAI LLM integration for LlamaIndex

(LlamaIndex-embeddings-openai) – OpenAI embeddings integration

chormadb - vector database

API Development :

FastAPI – Modern, fast web framework for building APIs

Uvicorn – ASGI server for running FastAPI applications

Technologies Used

AI & Machine Learning

Pydantic – Data validation and settings management using Python type annotations

Docker – Containerization platform (Dockerfiles present per service)

Data Processing & Analysis :

Pandas – Data manipulation and analysis

NumPy – Numerical computing

Matplotlib , WordCloud , Seaborn - Visualization

System & Monitoring :

Logging-config – Logging configuration utilities

Technologies Used

Data Analysis

Python: language used for data generation, processing, and analysis.

Regex (with re library): text cleaning, normalization and pattern matching.

Pandas: data collection, cleaning, transformation, and exploratory analysis.

Matplotlib & WordCloud: for data visualization.

Google colab

Execution Plan

Planning & Research

- Define requirements, architecture design, and user needs analysis.

Design Phase

- Create UI/UX designs.
- Map user flows and wireframes.

Development Phase

- Frontend with Flutter.
- Backend with FastAPI + Database.
- Integrate AI (OpenAI, LlamaIndex, ChromaDB).

Testing & QA

- Unit and integration testing.
- User experience testing.

Deployment

- Launch MVP using Docker + Cloud Hosting (Firebase / AWS).

Expected Impact

- **Students & Fresh Graduates**
Gain clear career roadmaps, mentorship, and instant technical support → better skills, higher confidence, and improved employability.
- **Mentors & Professionals**
Opportunity to share expertise, expand networks, and support the next generation of talent.
- **Universities & Institutions**
Enhanced student success rates, stronger industry alignment, and improved graduate outcomes.
- **Society & Economy**
Reduced skill gaps, stronger innovation culture, and contribution to economic growth through a more job-ready workforce.

Thank you!

